Maclaurin Series & Convergence of Series

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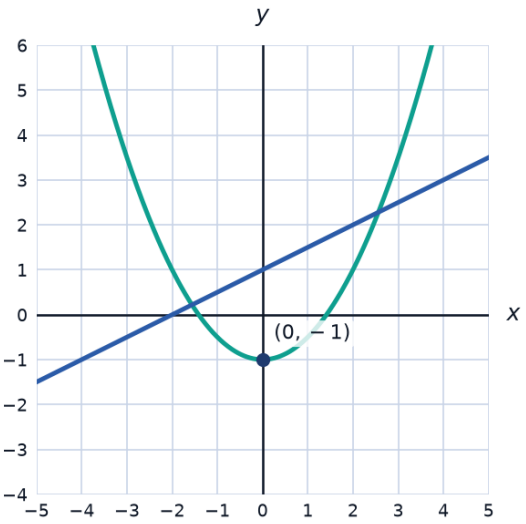
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Maclaurin Series Formula

Find the coefficient of x^2 in the Maclaurin series of e^{3x}.

2 [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of x^3 in the Maclaurin series of e^{3x}.



3 [Foundational] • Target: The Ratio Test for Convergence

Find the coefficient of x^3 in the Maclaurin series of $\sin(2x)$.

4 [Foundational] • Target: Radius and Interval of Convergence

Find the coefficient of x^4 in the Maclaurin series of $\cos(2x)$.

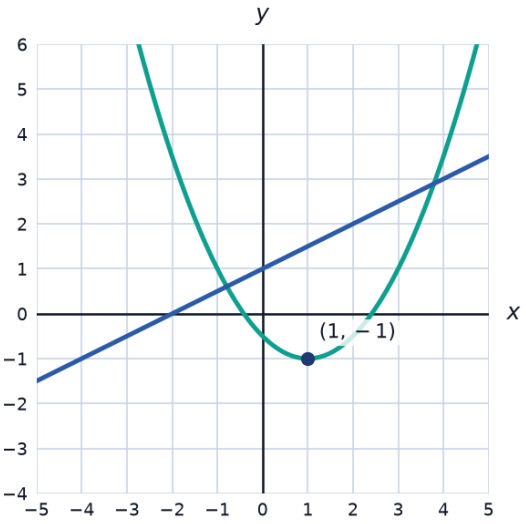
5 [Foundational] • Target: The Maclaurin Series Formula

Find the coefficient of x^3 in the Maclaurin series of $\sin(x)$.

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6 [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of x^4 in the Maclaurin series of $\cos(x)$.



7 [Foundational] • Target: The Ratio Test for Convergence

Find the coefficient of x^2 in the Maclaurin series of $\cos(x)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Radius and Interval of Convergence

Find the coefficient of x^2 in the Maclaurin series of $\sin(3x)$.

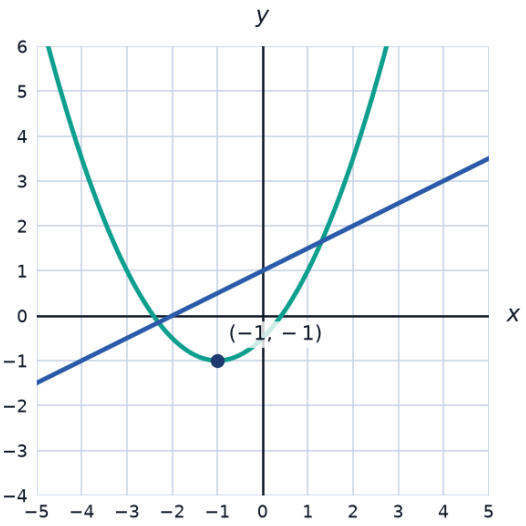
9 [Intermediate] • Target: The Maclaurin Series Formula

Find the coefficient of x^2 in the Maclaurin series of $\sin(2x)$.

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10 [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of x^4 in the Maclaurin series of $\sin(x)$.



11 [Intermediate] • Target: The Ratio Test for Convergence

Find the coefficient of x^2 in the Maclaurin series of $\sin(x)$.

12 [Intermediate] • Target: Radius and Interval of Convergence

Fourier series period 2π: $a_0 =$.

13 [Intermediate] • Target: The Maclaurin Series Formula

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 5$ cm.

14 [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

A sphere's radius increases at $\frac{dr}{dt} = 4$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 6$ cm.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

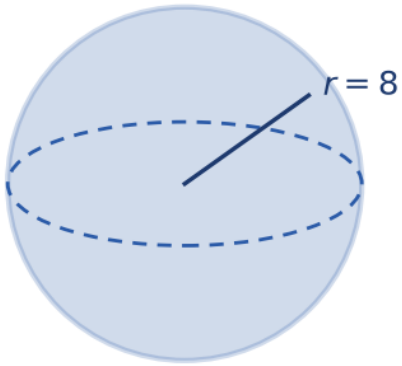
15 [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 4.02 The Poisson Distribution

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 5$ cm.

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16 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 5.01 Further Integration Techniques

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 8$ cm.



17 [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 5.02 Differential Equations

A sphere's radius increases at $\frac{dr}{dt} = 4$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 4$ cm.

18 [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A sphere's radius increases at $\frac{dr}{dt} = 2$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 4$ cm.

19 [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 6.01 Matrices (AI HL)

A sphere's radius increases at $\frac{dr}{dt} = 2$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 7$ cm.

20 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 7$ cm.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Maclaurin Series Formula

What is the general formula for the Maclaurin series expansion of a function $f(x)$?

a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$

b. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

c. $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

d. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

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22 [Foundational] • Target: The Maclaurin Series Formula

Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

a. $1 - 2x^2 + \frac{2}{3}x^4$

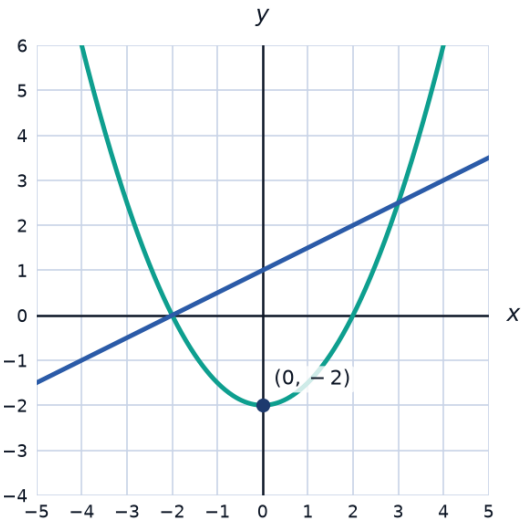
b. $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

c. $1 - 4x^2 + 16x^4$

d. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

23 [Foundational] • Target: The Maclaurin Series Formula

What is the coefficient of x^3 in the Maclaurin series of e^{-3x}?



a. $-\frac{9}{2}$

b. $\frac{9}{2}$

c. -9

d. -27

24 [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the Maclaurin series for $\ln(1 + 3x)$ up to the x^3 term.

a. $3x - \frac{9}{2}x^2 + 9x^3$

b. $3x + \frac{9}{2}x^2 + 9x^3$

c. $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

d. $3x - 9x^2 + 27x^3$

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25 [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Evaluate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

a. $\frac{1}{2}$

b. 1

c. 0

d. ∞

26 [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

a. $1 - x^2$

b. $1 + x^2$

c. $1 - x$

d. $x - x^3$

27 [Foundational] • Target: The Ratio Test for Convergence

What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n \to \infty} \frac{a_{n+1}}{a_n}$?

a. Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$

b. Converges if $L = 1$, diverges otherwise

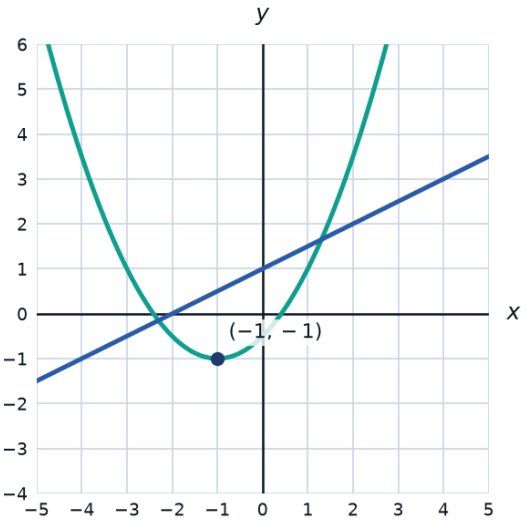
c. Diverges if $L < 1$, converges if $L > 1$

d. Always inconclusive for power series

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28 [Foundational] • Target: The Ratio Test for Convergence

Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.



a. Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$

b. Diverges, because $L = 3 > 1$

c. Inconclusive, because $L = 1$

d. Diverges, because the terms do not approach 0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Ratio Test for Convergence

What is the limit L in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. 1 (test is inconclusive)

b. 0

c. ∞

d. 1/2

30 [Intermediate] • Target: Radius and Interval of Convergence

Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

a. $R = \frac{1}{2}$

b. $R = 2$

c. $R = 1$

d. $R = \infty$

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31 [Intermediate] • Target: Radius and Interval of Convergence

What is the radius of convergence of the Maclaurin series for e^x?

a. $R = \infty$

b. $R = 1$

c. $R = e$

d. $R = 0$

32 [Intermediate] • Target: Radius and Interval of Convergence

For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

a. $(-1, 1)$

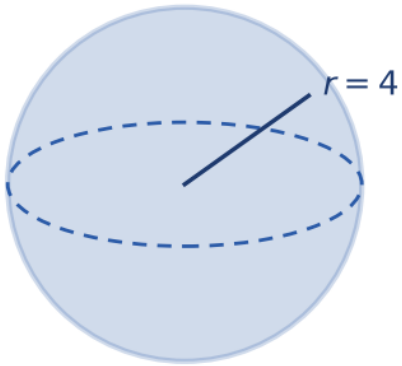
b. $[-1, 1]$

c. $(-1, 1]$

d. $[-1, 1)$

33 [Intermediate] • Target: The Maclaurin Series Formula

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of π) when $r = 4$ cm.



a. 24π cm^3/s

b. 74π cm^3/s

c. 64π cm^3/s

d. 44π cm^3/s

34 [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

In the expansion of $(1 + x)^{10}$, find the ratio of the coefficient of x^5 to the coefficient of x^4.

a. $\frac{8}{5}$

b. 1

c. $\frac{7}{5}$

d. $\frac{6}{5}$

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35 [Intermediate] • Target: The Ratio Test for Convergence

In the expansion of $(1+x)^{10}$, find the ratio of the coefficient of x^7 to the coefficient of x^6.

a. $\frac{4}{7}$

b. $\frac{3}{7}$

c. $\frac{5}{7}$

d. $\frac{6}{7}$

36 [Intermediate] • Target: Radius and Interval of Convergence

In the expansion of $(1+x)^{10}$, find the ratio of the coefficient of x^3 to the coefficient of x^2.

a. $\frac{8}{3}$

b. $\frac{10}{3}$

c. 3

d. $\frac{7}{3}$

37 [Intermediate] • Target: The Maclaurin Series Formula

In the expansion of $(1+x)^8$, find the ratio of the coefficient of x^6 to the coefficient of x^5.

a. $\frac{1}{2}$

b. 3

c. $\frac{3}{2}$

d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

In the expansion of $(1+x)^7$, find the ratio of the coefficient of x^6 to the coefficient of x^5.

a. 1

b. $\frac{1}{3}$

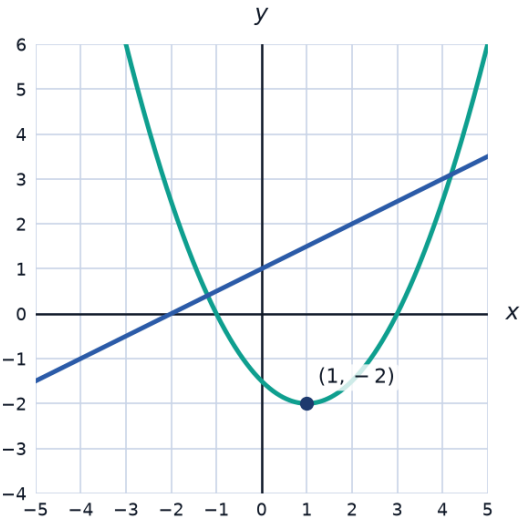
c. $\frac{2}{3}$

d. 2

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39 [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 6.01 Matrices (AI HL)

In the expansion of $(1 + x)^6$, find the ratio of the coefficient of x^4 to the coefficient of x^3.



a. $\frac{5}{4}$

b. $\frac{1}{2}$

c. 1

d. $\frac{3}{4}$

40 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

In the expansion of $(1 + x)^8$, find the ratio of the coefficient of x^5 to the coefficient of x^4.

a. $\frac{6}{5}$

b. $\frac{4}{5}$

c. 1

d. $\frac{3}{5}$

41 [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 6.03 Markov Chains (AI HL)

In the expansion of $(1 + x)^9$, find the ratio of the coefficient of x^8 to the coefficient of x^7.

a. $\frac{3}{4}$

b. $\frac{1}{4}$

c. $\frac{1}{2}$

d. $\frac{3}{2}$

42 [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 1.01 Proof by Mathematical Induction

In the expansion of $(1 + x)^7$, find the ratio of the coefficient of x^4 to the coefficient of x^3.

a. 1

b. 2

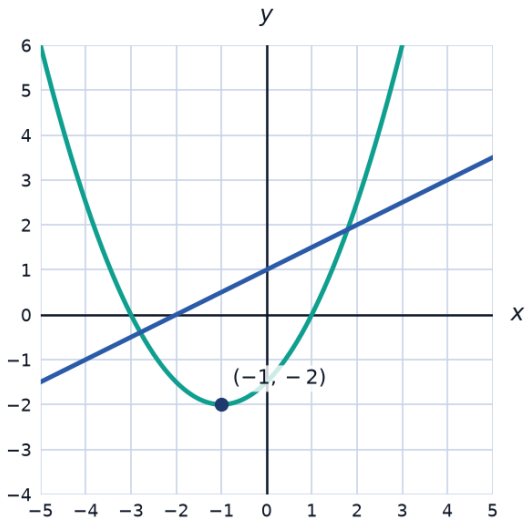
c. 3

d. 6

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43 [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

In the expansion of $(1 + x)^8$, find the ratio of the coefficient of x^3 to the coefficient of x^2.



a. 3

b. 4

c. 1

d. 2

44 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Reduction formula $I_n = \int_0^{\pi/2} \sin^n x\,dx$: $I_n = \frac{n-1}{n} I_{n-2}$. Find I_4:

a. $3\pi/8$

b. $3\pi/16$

c. $\pi/4$

d. $\pi/2$

45 [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 1.04 Partial Fractions

Let z_1 and z_2 be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where a and b are real numbers, find the value of $a^2 + b^2$.

a. $\frac{25}{13}$

b. $\frac{5}{13}$

c. $\frac{169}{25}$

d. $\frac{13}{5}$

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